

Zeju Li

 [GitHub](#) |  [Google Scholar](#) |  [Homepage](#) |  [E-mail](#)

EDUCATION

Zhejiang University

Hangzhou, Zhejiang, China

B.Eng. in Computer Science and Technology

- College of Computer Science and Technology
- Mixed Class, [Chu Kochen Honors College](#)

Sep 2025 – Jun 2027 (expected)

Sep 2023 – Jun 2025

RESEARCH INTEREST

I'm passionate about developing truly intelligent robots capable of general-purpose autonomy in human-centric environments. My research centers on world models, robot foundation models, and agentic architectures that enable robots to generalize across tasks and learn efficiently through interaction. I am particularly interested in unifying of perception, locomotion, and dexterous manipulation within a single embodied agent as a step toward general intelligence grounded in the physical world.

RESEARCH EXPERIENCE

Northwestern University

Research Intern at [MLL-Lab](#)

Feb 2026 – Present

Advised by [Prof. Manling Li](#) and [Prof. Ruohan Zhang](#)

Evaluating Latency Sensitivity in Real-Time Foundation Model Agents

Feb 2026 – Jul 2026

- Reframed inference latency as a first-class dimension of real-time agent evaluation rather than a hardware artifact.
- Established a general framework for measuring how foundation-model agents respond to train-deployment timing mismatch.
- Revealed that latency robustness is task-dependent: training can mitigate brittleness, but current models do not eliminate it.

Zhejiang University

Hangzhou, Zhejiang, China

Research Assistant at [State Key Lab of CAD&CG](#)

Oct 2024 – Present

Advised by [Prof. Hao Chen](#) and [Prof. Chunhua Shen](#)

Learning Robust Alignment from Critical Motion Segments

Mar 2026 – Jun 2026

- Co-led the study showing that fluent demonstrations underrepresent critical alignment and recovery states in fine-grained manipulation.
- Identified a supervision mismatch: critical alignment and recovery are compressed into brief motion segments and underrepresented during imitation learning.
- Proposed STAIR, a compact spatiotemporal interface that turns local motion into dense supervision and conditions vision-language-action policies on short-horizon motion.

Reconstructing, Generating, and Retargeting Hand-Object Interactions

Sep 2025 – Nov 2025

- Investigated how reconstructed human hand-object interactions can be transferred into simulation as dexterous-manipulation demonstrations.
- Built a real-to-sim pipeline that retargets MANO hand trajectories to a dexterous robotic hand and imports reconstructed object assets into Isaac Gym, validating AGILE’s simulation readiness.

Compact State Representations for Efficient World Modeling and Action Learning

Jul 2025 – Sep 2025

- Co-developed the hypothesis that robot observations can be compressed into a highly compact token space in which motion emerges as differences between states.
- Led the study on bimanual manipulation data, examining whether severe compression can preserve the structure underlying coordinated actions.
- Showed through targeted representation probing that compact state differences retain linearly decodable action information, supporting a unified representation of state and motion.

Language-Guided Long-Horizon Mobile Manipulation

Jun 2025 – Aug 2025

- Contributed to a unified framework integrating hierarchical vision-language planning with whole-body control for long-horizon quadruped manipulation in open-world environments.
- Led the construction of its long-horizon benchmark, designing compositional tasks and evaluation protocols and establishing baseline performance across diverse indoor and outdoor scenarios.

PUBLICATIONS

* EQUAL CONTRIBUTIONS, † CORRESPONDING AUTHOR

[1] Perfect Demo Makes Poor Teacher: Learning Robust Alignment from Critical Motion Segments

Mingyu Liu^{*}, Zeju Li^{*}, Jiuhe Shu, Hanqing Wang, Yuhao Chao, Hao Chen[†], Chunhua Shen[†]
Preprint, 2026 [\[Paper\]](#) [\[Project Page\]](#)

[2] StaMo: Unsupervised Learning of Generalizable Robot Motion from Compact State Representation

Mingyu Liu^{*}, Jiuhe Shu^{*}, Hui Chen, Zeju Li, Canyu Zhao, Jiange Yang, Shenyuan Gao, Hao Chen[†], Chunhua Shen[†]
CVPR (Highlight Presentation, Top 3.1%), 2026 [\[Paper\]](#) [\[Project Page\]](#)

[3] ODYSSEY: Open-World Quadrupeds Exploration and Manipulation for Long-Horizon Tasks

Kaijun Wang^{*}, Liqin Lu^{*}, Mingyu Liu, Jianuo Jiang, Zeju Li, Bolin Zhang, Wancai Zheng, Xinyi Yu, Hao Chen[†], Chunhua Shen[†]
AAAI (Oral Presentation, Top 3.5%), 2026 [\[Paper\]](#) [\[Project Page\]](#)

[4] AGILE: Hand-Object Interaction Reconstruction from Video via Agentic Generation

JinChuan Shi^{*}, Binhong Ye^{*}, Tao Liu, Junzhe He, Yangjinhui Xu, Xiaoyang Liu, Zeju Li, Hao Chen[†], Chunhua Shen[†]
SIGGRAPH (Conference Track), 2026 [\[Paper\]](#) [\[Project Page\]](#)

SKILLS

Programming Languages Python (PyTorch), C/C++, L^AT_EX, SQL, Linux
 Chinese (Native), English (TOEFL 100/120, Sep 2025)